Computer Communication & Networks

Lecture 1

Introduction

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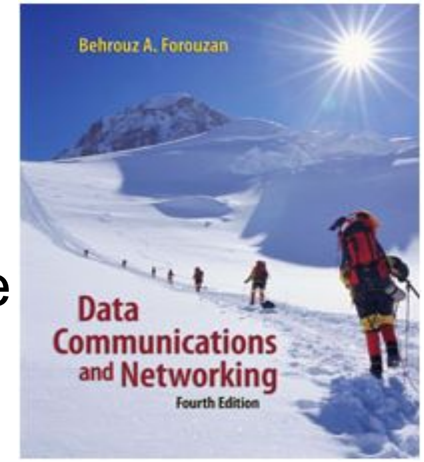
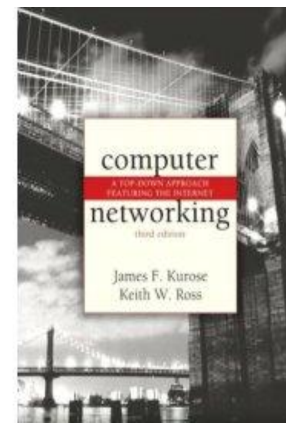
Reading

- Text book:
  - Data Communications and Networking, 4/e
    - B.A. Forouzan,
      McGraw-Hill, 2003,
      ISBN 0-07-292354-7.



- Reference books:
  - Computer Networking, a top-down approach featuring the Internet (3rd edition),
    - J.K.Kurose, K.W.Ross,
      Addison-Wesley, 2005,
      ISBN 0-321-26976-4.
  - Computer Networks, A Systems Approach

    L. Peterson & Davie



Required Skills

- The course does not assume prior knowledge of networking.

My Requirement from YOU

- I require YOU to take active part during lectures
  - Which means Lot of Questioning in the class – (Interactive session)

Aim of the Course

- Aim of the course is to introduce you to the world of computer networks, so that you could
  - know the science being used in running this network
  - Use this knowledge in your professional field

Network design

Before looking inside a computer network, first agree on **what** a computer network is

Computer network ?

- Set of serial lines to attach terminals to mainframe ?
- Telephone network carrying voice traffic ?
- Cable network to disseminate video signals ?

Specialized to handle:

Keystrokes

Voice

Video

What distinguishes a Computer network ?

- Generality
- Built from general purpose programmable hardware
- Supports wide range of applications

Information, Computers, Networks

- Information: anything that is represented in *bits*
  - Form (can be represented as bits) vs
  - Substance (cannot be represented as bits)
- Properties:
  - Infinitely replicable
  - Computers can "***manipulate***" information
  - Networks create "***access***" to information

Networks

- Potential of networking:
  - move bits everywhere, cheaply, and with desired performance characteristics
- Network provides "connectivity"

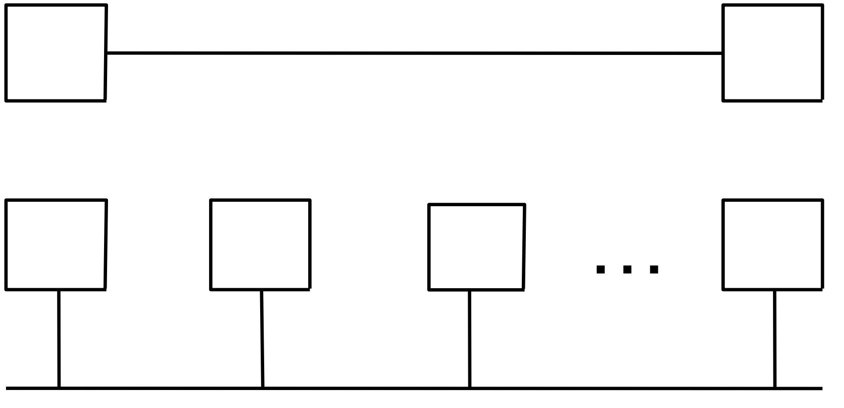
What is "Connectivity" ?

- Direct or indirect access to every other node in the network

- Connectivity is the magic needed to communicate if you do not have a direct pt-pt physical link.
  - *Tradeoff:* *Performance characteristics worse than true physical link!*

Building Blocks

- Nodes: PC, special-purpose hardware…
  - hosts
  - switches
- Links: coax cable, optical fiber…
  - point-to-point
  - multiple access

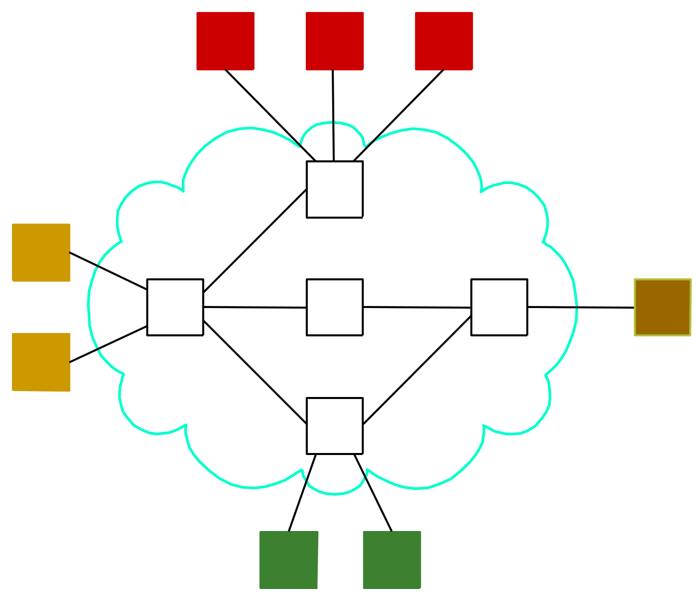


Why not connect each node with every other node ?

- Number of computers that can be connected becomes very **limited**
- Number of wires coming out of each node becomes **unmanageable**
- Amount of physical hardware/devices required becomes very **expensive**
- Solution: indirect connectivity using intermediate data forwarding nodes

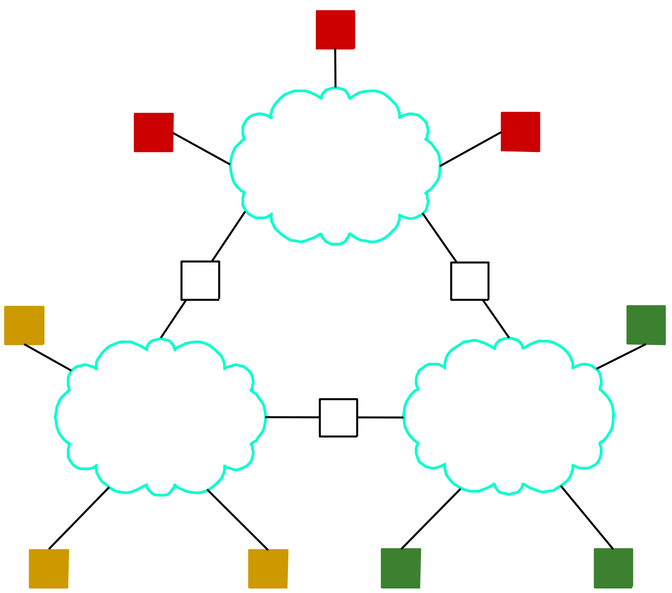
Switched Networks

- A network can be defined recursively as...
  - two or more nodes connected by a link
  - white nodes (switches) ***implement*** the network
  - colored nodes (hosts) ***use*** the network



Switched Networks

- A network can be defined recursively as...
  - two or more networks connected by one or more nodes: internetworks
  - white nodes (router or gateway) ***interconnects*** the networks
  - a ***cloud*** denotes "any type of independent network"



A Network

A network can be defined recursively as

two or more nodes connected by a physical link

Or

two or more networks connected by one or more nodes

Switching Strategies

- Circuit switching: carry bit streams
  a. establishes a dedicated circuit
  b. links reserved for use by communication channel
  c. send/receive bit stream at constant rate
  d. example: original telephone network

- Packet switching: store-and-forward messages
  a. operates on discrete blocks of data
  b. utilizes resources according to traffic demand
  c. send/receive messages at variable rate
  d. example: Internet

What next ?

- Hosts are directly or indirectly connected to each other
  - Can we now provide host-host connectivity ?
- Nodes must be able to say **which** host it wants to communicate with

Addressing and Routing

- Address: byte-string that identifies a node
  - usually unique
- Routing: forwarding decisions
  - process of determining how to forward messages to the destination node based on its address
- Types of addresses
  - unicast: node-specific
  - broadcast: all nodes on the network
  - multicast: some subset of nodes on the network

Wrap-up

- A network can be constructed from ***nesting*** of networks

- An ***address*** is required for each node that is reachable on the network

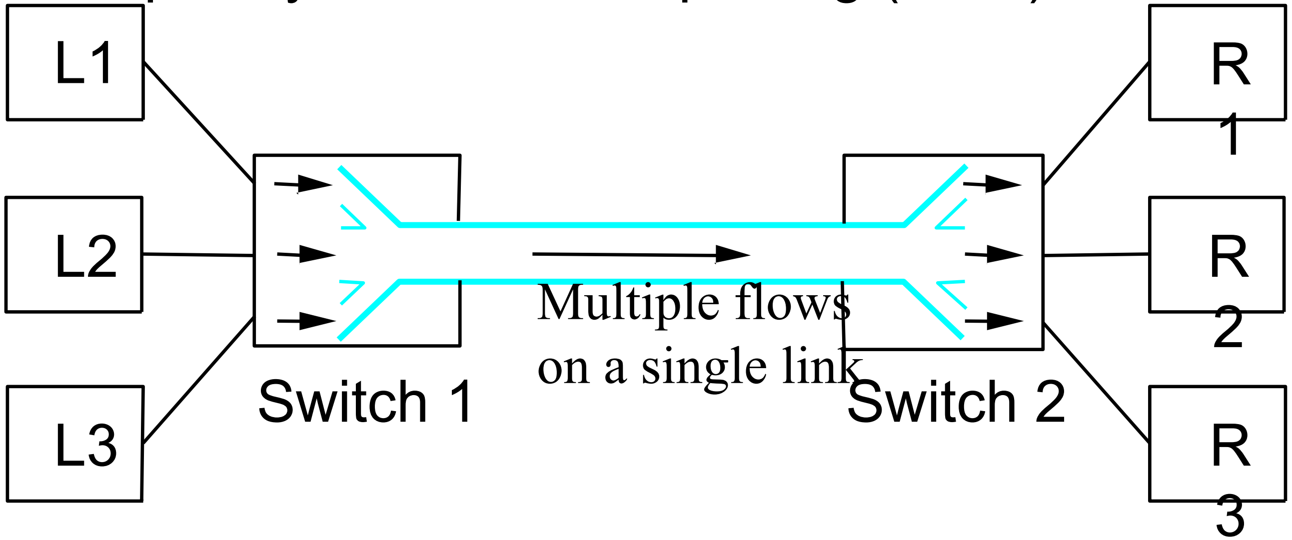
- Address is used to ***route*** messages toward appropriate destination

What next ?

- Hosts know how to reach other hosts on the network
- How should a node use the network for its communication ?
- All pairs of hosts should have the ability to exchange messages: **cost-effective** resource sharing for efficiency

Multiplexing

- Physical links and nodes are shared among users
  - (synchronous) Time-Division Multiplexing (TDM)
  - Frequency-Division Multiplexing (FDM)



Do you see any problem with TDM / FDM ?

What Goes Wrong in the Network?

Reliability at stake

- Bit-level errors (electrical interference)
- Packet-level errors (congestion)
  - distinction between lost and late packet
- Link and node failures
  - distinction between broken and flaky link
  - distinction between failed and slow node

What Goes Undesirable in the Network?

Required performance at stake

- Messages are delayed
- Messages are delivered out-of-order
- Third parties eavesdrop

- The challenge is to fill the gap between application expectations and hardware capabilities

Research areas in Networking

- Routing
- Security
- Ad-hoc networks
- Wireless networks
- Protocols
- Quality of Service
- …

Readings

- Chapter 1: 1.1, 1.2
  - Computer Networks, A Systems Approach
    L. Peterson & Davie